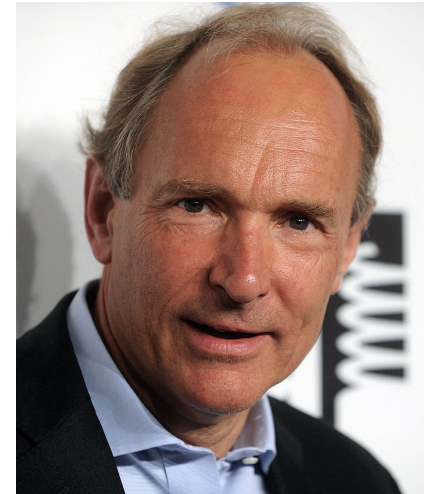


Web Accessibility

Megan Kwok & Vincent Lam

What is Accessibility

- Designing websites that include considerations for visual, auditory, motor, and cognitive impairments
 - Inclusivity & upholds universality of the web
 - “The power of the Web is in its universality. Access by everyone regardless of disability is an essential aspect.” - Tim Berners-Lee (Director of W3C & Inventor of WWW)
 - Equality of opportunity, autonomy, and independence
 - Prevents discrimination
 - Legal compliance
 - Search Engine Optimization (SEO) benefits
 - Reputation



WCAG

- WCAG: Web Content Accessibility Guidelines
 - Created by W3C (World Wide Web Consortium)
 - Three Levels of Compliance: A, AA, AAA
 - Commonly used as benchmark in ADA court cases & explicitly mandated in EU Web Accessibility Directive
 - 87 criteria in latest version



WCAG

- **Perceivable:** Information must be presented so users can perceive it
 - Alt text for images
 - Vid captions
 - Proper color contrast for readability.
 - Scalable text and adaptable layouts.
- **Operable:** Users must be able to navigate and interact:
 - Keyboard accessibility (no mouse required).
 - Clear and consistent navigation.
 - Enough time to read or interact with content.
 - Avoiding content that causes seizures (like flashing lights).
- **Understandable:** Content and interface must make sense:
 - Clear language/instructions
 - Consistent layout and predictable functionality.
 - Helpful error messages and guidance for forms.
- **Robust:** Content must work across devices and assistive technologies:
 - Compatible with screen readers, magnifiers, voice control, etc.
 - Follows coding standards to ensure future compatibility.

Auditing

- Automated, systematic check of a website for accessibility barriers.
- Compares site features against WCAG standards.
- Identifies issues in perception, navigation, content clarity, and technology compatibility.
- Provides actionable recommendations to make the site usable for everyone.
 - E.g. tabnav.com (AI-based service)
- Audit Report on stuy.enschool.org
 - 54% compliant
 - 113 Violations
 - 61% Visual Impairments
 - 67% Motor Impairments
 - 75% Cognitive Impairments
 - 50% Other Impairments

WE CAN BE SUED!!!

Stuy Website Changes

- Titles built as text tags should be labeled as headings for assistive technology
 - Assistive Technologies use headings to navigate the page

```
<span class="heading">Clubs and Activities</span>
```

```
<span class="heading"><h3>Clubs and Activities</h3></span>
```

- Pages must have proper landmark structure (one main, one banner, one footer)
 - Multiple landmarks means users have to navigate each one to find info

```
Missing landmarks: main
```

```
<main></main> distinguisher added
```

- SVG and Icon elements must include ARIA tags (next slide)

```
<svg width="24" height="24" viewBox="0 0 24 24" focusable="false"
class="VlpgJd-yAWNEb-hvhgNd-THI6Vb NMm5M">
```

```
<svg width="24" height="24" viewBox="0 0 24 24" focusable="false"
class="VlpgJd-yAWNEb-hvhgNd-THI6Vb NMm5M" aria-hidden="true">
```

- Page should only have one heading
 - Multiple page headings cause ambiguity about page content

```
<h1>Pro Scientia Atque Sapientia</h1>
<h1><a href="/apps/news/" class="news-main-title">News
<span>&amp;</span> Announcements</a></h1>
```

```
<h1>Pro Scientia Atque Sapientia</h1>
<h2><a href="/apps/news/" class="news-main-title">News <span>&amp;</span>
Announcements</a></h2>
```

- Interactive controls must have appropriate semantic roles

```
"<div onclick=""_enSearchPopupToggle()"" tabindex=""0""
class=""search-wrapper all-icon ""> <img src=""/
pics/search_button.png"" alt=""Search Button""> </div>"
```

```
"<div onclick=""_enSearchPopupToggle()"" tabindex=""0"" class=""search-wrapper
all-icon ""> <img src=""/
pics/search_button.png"" alt=""Search Button"" role=""button"> </div>"
```

What are SVGs?

SVGs: code based



whose coding (due to W. ICKING) is;

```
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\loffset{0.5}\{ibbu0f9}\roffset{0.5}\{tbu0}\hu j\en
```

Raster Imgs (.jpg, .png, .jpeg): binary based

/home/tilman/PNG-Gradient.png - Bless

File Edit View Search Tools Help

PNG-Gradient.png x

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00000022	00	00	C2	49	44	41	54	78	5E	ED	D4	81	06	C3	30	14	40	IDATx^.....0.@
00000033	D1	B7	34	DD	FF	FF	6F	B3	74	56	EA	89	12	6C	28	73	E2	..4...o.tV...l(s.
00000044	AA	34	49	03	87	D6	FE	D8	7B	89	BB	52	8D	3B	87	FE	01	.4I.....{.R.;...
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00000088	D1	F4	92	5C	5C	3E	CF	9C	3F	73	71	58	5F	AF	8B	79	5B	...\\>..?sqX...y[
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Unsigned 8 bit:	13	Unsigned 32 bit:	218765834	Decimal:	013 010 026 010	
Signed 16 bit:	3338	Float 32 bit:	4,255588E-31	Octal:	015 012 032 012	
Unsigned 16 bit:	3338	Float 64 bit:	7,46625117388175E-246	Binary:	00001101 00001010 00	
☐ Show little endian decoding		☐ Show unsigned as hexadecimal		ASCII Text: 0D 0A 1A 0A		
Offset: 0x4 / 0xfa				Selection: 0x1 to 0x3 (0x3 bytes)		INS

Our Accessibility Checker

- Automated accessibility auditor using regex
- Checks for specific WCAG guidelines
 - Title tag labels
 - Landmark structure
 - ARIA Tags for SVG and Icons
 - No more than one main heading
- Returns non-compliant lines and displays reason for violation

Our Website

- Line 70: Missing valid <h1>-<h6>: our favorite

```
<span class="heading">our favorite class:</span>
```

```
<span class="heading"><h3>our favorite  
class:</h3></span>
```

- Missing tags: header, main

Missing tags header, main

header and main tags added

- Found 2 <h1> tags. There should only be one per page.

```
<h1 style="text-align: center;">We are  
KWAM</h1>  
  <h1 style="text-align: center;">us and our  
duckies</h1>
```

```
<h1 style="text-align: center;">We are KWAM</h1>  
  <h2 style="text-align: center;">us and our  
duckies</h2>
```

- Interactive control must have semantic role: ('div', 'onclick')

```
<div id="colorChangeButton"  
onclick="changeColor();">  
  <p>click this!</p>  
</div>
```

```
<div id="colorChangeButton" onclick="changeColor();" role="button">  
  <p>click this!</p>  
</div>
```

Citations

- 1) <https://www.w3.org/WAI/WCAG22/quickref/>
- 2) <https://www.sciencedirect.com/science/article/pii/S1877050923002673>
- 3) https://link.springer.com/chapter/10.1007/978-3-319-22629-3_16
- 4) <https://dl.acm.org/doi/full/10.1145/3763243>
- 5) <https://www.ada.gov/resources/web-guidance/>